

Completing the square when a is not 1

9709 PURE 1 · 1.1 QUADRATICS · TWO ROUTES, ONE ANSWER

Before you start, look for the equals sign. That one thing picks the route.

There is an = 0. “solve” · “find x” · “find the roots” · “find k for a repeated root”

divide by a

There is no equals sign. “write in the form $A(x + B)^2 + C$ ” · “find the vertex” · “find the minimum value”

factor a out

1 · There is an equals sign, so divide

$$3x^2 + 4x - 15 = 0$$

ROUTE 2 · DIVIDE BY A ← USE THIS ONE

Divide every term by 3. The other side is 0, and $0 \div 3$ is still 0.

$$x^2 + 4/3 x - 5 = 0$$

Move the -5 across.

$$x^2 + 4/3 x = 5$$

Half of $4/3$ is $2/3$. Square it, $4/9$, and add it to **both** sides.

$$(x + 2/3)^2 = 5 + 4/9 = 49/9$$

Square root both sides.

$$x + 2/3 = \pm 7/3 \rightarrow x = 5/3 \text{ or } x = -3$$

ROUTE 1 · FACTOR A OUT ALSO CORRECT, JUST LONGER

Pull the 3 out of the x^2 and x terms.

$$3(x^2 + 4/3 x) - 15 = 0$$

Half of $4/3$ is $2/3$. Subtract $4/9$ **inside** the bracket.

$$3[(x + 2/3)^2 - 4/9] - 15 = 0$$

Multiply back out. The $4/9$ is inside, so it becomes $4/3$.

$$3(x + 2/3)^2 - 49/3 = 0$$

Add $49/3$, divide by 3, square root.

$$(x + 2/3)^2 = 49/9 \rightarrow x = 5/3 \text{ or } x = -3$$

2 · No equals sign, so factor out

$$2x^2 - 12x + 5$$

Write in the form $A(x + B)^2 + C$. This is an identity, true for every x , so there is no other side to divide into. Setting it equal to zero yourself does not help. That solves for x instead of rewriting the expression.

ROUTE 1 · FACTOR A OUT ← USE THIS ONE

$$2(x^2 - 6x) + 5$$

$$2[(x - 3)^2 - 9] + 5$$

$$2(x - 3)^2 - 18 + 5$$

$$2(x - 3)^2 - 13$$

ROUTE 2 · DIVIDE, THEN PUT THE A BACK

$$x^2 - 6x + 5/2$$

$$(x - 3)^2 - 9 + 5/2$$

$$(x - 3)^2 - 13/2 \text{ not finished}$$

$$\times 2 \rightarrow 2(x - 3)^2 - 13$$

The trap, and it is the same trap on both routes: the a

Route 1 loses it by bringing the -9 back out as -9 instead of **-18**, which gives $2(x - 3)^2 - 4$. **Route 2 loses it** by stopping one line early at $(x - 3)^2 - 13/2$.

The bracket is right either way, so the line of symmetry $x = 3$ survives both mistakes. The minimum value does not. It is **-13**, not -4 and not $-13/2$. If a question asks for the minimum, the maximum, the vertex or the range, that number is the answer, so the a has to be back before you put your pen down.